

[Github link](#)

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WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:8002
* Running on http://172.18.0.2:8002
Press CTRL+C to quit
172.18.0.4 - - [22/Feb/2026 19:02:26] "POST /counter HTTP/1.1" 200 -
172.18.0.4 - - [22/Feb/2026 19:02:32] "POST /counter HTTP/1.1" 200 -
172.18.0.4 - - [22/Feb/2026 19:02:39] "POST /counter HTTP/1.1" 200 -
172.18.0.4 - - [22/Feb/2026 19:02:45] "POST /counter HTTP/1.1" 200 -
172.18.0.4 - - [22/Feb/2026 19:02:51] "POST /counter HTTP/1.1" 200 -
172.18.0.4 - - [22/Feb/2026 19:03:05] "GET /counter/alice HTTP/1.1" 200 -
172.18.0.4 - - [22/Feb/2026 19:03:13] "GET /counter/bob HTTP/1.1" 200 -
172.18.0.4 - - [22/Feb/2026 19:03:20] "GET /counter HTTP/1.1" 200 -
* Serving Flask app 'main'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:8000
* Running on http://172.18.0.4:8000
Press CTRL+C to quit
192.168.65.1 - - [22/Feb/2026 19:02:26] "POST /transaction HTTP/1.1" 200 -
192.168.65.1 - - [22/Feb/2026 19:02:32] "POST /transaction HTTP/1.1" 200 -
192.168.65.1 - - [22/Feb/2026 19:02:39] "POST /transaction HTTP/1.1" 200 -
192.168.65.1 - - [22/Feb/2026 19:02:45] "POST /transaction HTTP/1.1" 200 -
192.168.65.1 - - [22/Feb/2026 19:02:51] "POST /transaction HTTP/1.1" 200 -
192.168.65.1 - - [22/Feb/2026 19:03:05] "GET /user/alice HTTP/1.1" 200 -
192.168.65.1 - - [22/Feb/2026 19:03:13] "GET /user/bob HTTP/1.1" 200 -
192.168.65.1 - - [22/Feb/2026 19:03:20] "GET /accounts HTTP/1.1" 200 -
192.168.65.1 - - [22/Feb/2026 19:03:28] "GET /stats HTTP/1.1" 200 -
192.168.65.1 - - [22/Feb/2026 19:03:45] "DELETE /stats HTTP/1.1" 200 -

• (base) sofiapopeniuk@MacBookPro architectureOfComputerSystems % curl -X POST http://localhost:8000/transaction \
  -H "Content-Type: application/json" \
  -d '{"user_id": "alice", "amount": 100}'
{"balance":100.0,"transaction_id":"256ce065-1029-44f4-a093-e1443c514a18"}
• (base) sofiapopeniuk@MacBookPro architectureOfComputerSystems % curl -X POST http://localhost:8000/transaction \
  -H "Content-Type: application/json" \
  -d '{"user_id": "alice", "amount": -30}'
{"balance":70.0,"transaction_id":"10ba1d1c-6a2e-4be7-a537-f5feba71deb5"}
• (base) sofiapopeniuk@MacBookPro architectureOfComputerSystems % curl -X POST http://localhost:8000/transaction \
  -H "Content-Type: application/json" \
  -d '{"user_id": "bob", "amount": 200}'
{"balance":200.0,"transaction_id":"cc9e0bd8-b028-45eb-bd4b-4363093c9b19"}
• (base) sofiapopeniuk@MacBookPro architectureOfComputerSystems % curl -X POST http://localhost:8000/transaction \
  -H "Content-Type: application/json" \
  -d '{"user_id": "alice", "amount": 50}'
{"balance":120.0,"transaction_id":"bb46ca22-094d-4788-80e8-af8af811c9a1"}
• (base) sofiapopeniuk@MacBookPro architectureOfComputerSystems % curl -X POST http://localhost:8000/transaction \
  -H "Content-Type: application/json" \
  -d '{"user_id": "bob", "amount": -20}'
{"balance":100.0,"transaction_id":"2542626b-4adc-4624-b7da-14b069fd1b82"}
• (base) sofiapopeniuk@MacBookPro architectureOfComputerSystems % curl http://localhost:8000/user/alice
{"balance":120.0,"transactions":[{"amount":100.0,"timestamp":1771786946.4915712,"transaction_id":"256ce065-1029-44f4-a093-e1443c514a18","user_id":"alice"},{"amount":-30.0,"timestamp":1771786952.5911734,"transaction_id":"10ba1d1c-6a2e-4be7-a537-f5feba71deb5","user_id":"alice"},{"amount":50.0,"timestamp":1771786965.7689238,"transaction_id":"bb46ca22-094d-4788-80e8-af8af811c9a1","user_id":"alice"}],"user_id":"alice"}
• (base) sofiapopeniuk@MacBookPro architectureOfComputerSystems % curl http://localhost:8000/user/bob
{"balance":100.0,"transactions":[{"amount":200.0,"timestamp":1771786959.2628987,"transaction_id":"cc9e0bd8-b028-45eb-bd4b-4363093c9b19","user_id":"bob"},{"amount":-20.0,"timestamp":1771786971.3148963,"transaction_id":"2542626b-4adc-4624-b7da-14b069fd1b82","user_id":"bob"}],"user_id":"bob"}
• (base) sofiapopeniuk@MacBookPro architectureOfComputerSystems % curl http://localhost:8000/accounts
{"balances":{"alice":120.0,"bob":100.0}}
• (base) sofiapopeniuk@MacBookPro architectureOfComputerSystems % curl http://localhost:8000/stats
{"counter_service":{"avg_ms":4.26,"total_calls":8,"total_ms":34.06},"logging_service":{"avg_ms":5.35,"total_calls":7,"total_ms":37.46}}
• (base) sofiapopeniuk@MacBookPro architectureOfComputerSystems % curl -X DELETE http://localhost:8000/stats
{"status":"stats reset"}
• (base) sofiapopeniuk@MacBookPro architectureOfComputerSystems % docker logs logging-service
docker logs counter-service
docker logs facade-service
* Serving Flask app 'main'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:8001
* Running on http://172.18.0.3:8001
Press CTRL+C to quit
172.18.0.4 - - [22/Feb/2026 19:02:26] "POST /log HTTP/1.1" 200 -
172.18.0.4 - - [22/Feb/2026 19:02:32] "POST /log HTTP/1.1" 200 -
172.18.0.4 - - [22/Feb/2026 19:02:39] "POST /log HTTP/1.1" 200 -
172.18.0.4 - - [22/Feb/2026 19:02:45] "POST /log HTTP/1.1" 200 -
172.18.0.4 - - [22/Feb/2026 19:02:51] "POST /log HTTP/1.1" 200 -
172.18.0.4 - - [22/Feb/2026 19:03:05] "GET /log/alice HTTP/1.1" 200 -
172.18.0.4 - - [22/Feb/2026 19:03:13] "GET /log/bob HTTP/1.1" 200 -
* Serving Flask app 'main'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:8002
```

Docker logs are too large for the tests below to screenshot them, so I don't insert them

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● (base) sofiiapopeniuk@MacBookPro architectureOfComputerSystems % python3 perf_test.py --scenario 1

=====
SCENARIO 1 | 10 clients x 10,000 requests each
=====
Each client credits their OWN account. Expected balance per account: 10,000

--- Results ---
user_1      ok= 10000 errors= 0 time=407.44s
user_2      ok= 10000 errors= 0 time=407.55s
user_3      ok= 10000 errors= 0 time=407.67s
user_4      ok= 10000 errors= 0 time=407.57s
user_5      ok= 10000 errors= 0 time=407.49s
user_6      ok= 10000 errors= 0 time=405.71s
user_7      ok= 10000 errors= 0 time=407.77s
user_8      ok= 10000 errors= 0 time=406.93s
user_9      ok= 10000 errors= 0 time=407.25s
user_10     ok= 10000 errors= 0 time=407.33s

--- Summary ---
Total requests : 100,000
Successful     : 100,000
Errors        : 0
Wall time     : 407.77s
Throughput    : 245 req/s

--- Service timing (accumulated) ---
logging-service : 100,000 calls total=1329161ms avg=13.29ms/call
counter-service : 100,000 calls total=1293401ms avg=12.93ms/call

--- Final balances ---
alice       balance = 120
bob         balance = 180
user_1      balance = 10,000
user_10     balance = 10,000
user_2      balance = 10,000
user_3      balance = 10,000
user_4      balance = 10,000
user_5      balance = 10,000
user_6      balance = 10,000
user_7      balance = 10,000
user_8      balance = 10,000
user_9      balance = 10,000
● (base) sofiiapopeniuk@MacBookPro architectureOfComputerSystems % curl http://localhost:8000/accounts
{"balances":{"alice":120.0,"bob":180.0,"user_1":10000.0,"user_10":10000.0,"user_2":10000.0,"user_3":10000.0,"user_4":10000.0,"user_5":10000.0,"user_6":10000.0,"user_7":10000.0,"user_8":10000.0,"user_9":10000.0}}
● (base) sofiiapopeniuk@MacBookPro architectureOfComputerSystems % curl http://localhost:8000/stats
{"counter_service":{"avg_ms":12.93,"total_calls":100002,"total_ms":1293409.76},"logging_service":{"avg_ms":13.29,"total_calls":100000,"total_ms":1329161.34}}

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● (base) sofiiapopeniuk@MacBookPro architectureOfComputerSystems % docker compose restart
[+] Restarting 3/3
✓ Container logging-service Started 11.1s
✓ Container facade-service Started 11.0s
✓ Container counter-service Started 11.0s
● (base) sofiiapopeniuk@MacBookPro architectureOfComputerSystems % curl -X DELETE http://localhost:8000/stats
{"status":"stats reset"}
● (base) sofiiapopeniuk@MacBookPro architectureOfComputerSystems % python3 perf_test.py --scenario 2

=====
SCENARIO 2 | 10 clients x 10,000 requests each
=====
All clients credit the SAME account. Expected balance: 100,000

--- Results ---
shared_user ok= 10000 errors= 0 time=437.90s
shared_user ok= 10000 errors= 0 time=438.13s
shared_user ok= 10000 errors= 0 time=437.68s
shared_user ok= 10000 errors= 0 time=438.14s
shared_user ok= 10000 errors= 0 time=437.22s
shared_user ok= 10000 errors= 0 time=436.94s
shared_user ok= 10000 errors= 0 time=437.73s
shared_user ok= 10000 errors= 0 time=438.19s
shared_user ok= 10000 errors= 0 time=437.80s
shared_user ok= 10000 errors= 0 time=437.72s

--- Summary ---
Total requests : 100,000
Successful     : 100,000
Errors        : 0
Wall time     : 438.19s
Throughput    : 228 req/s

--- Service timing (accumulated) ---
logging-service : 100,000 calls total=1368713ms avg=13.69ms/call
counter-service : 100,000 calls total=1330053ms avg=13.30ms/call

--- Final balances ---
shared_user balance = 100,000
● (base) sofiiapopeniuk@MacBookPro architectureOfComputerSystems % curl http://localhost:8000/accounts
{"balances":{"shared_user":100000.0}}
● (base) sofiiapopeniuk@MacBookPro architectureOfComputerSystems % curl http://localhost:8000/stats
{"counter_service":{"avg_ms":13.3,"total_calls":100002,"total_ms":1330075.07},"logging_service":{"avg_ms":13.69,"total_calls":100000,"total_ms":1368713.05}}

```